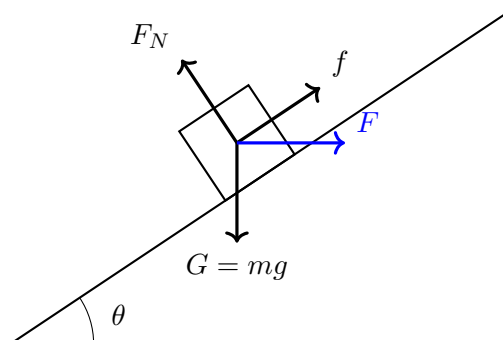

Q July 26, 2026
Holding a Block on an Incline

JASON ZHANG, COLLEGEBOARD

[7 pts]

A block of mass m rests on a rough incline that makes an angle θ with the horizontal. The coefficient of static friction between the block and the incline is μ .



An additional horizontal force F is applied to the block, parallel to the ground (positive F taken to the right; F may be negative, i.e. directed to the left). Find the range of values of F for which the block remains stationary.

Assumptions

1. The incline is fixed; the block does not tip.
2. Static friction can take any value between $-\mu N$ and $+\mu N$ along the incline.
3. The applied force F is horizontal, parallel to the ground; positive F points to the right, and a negative value of F means it points to the left.